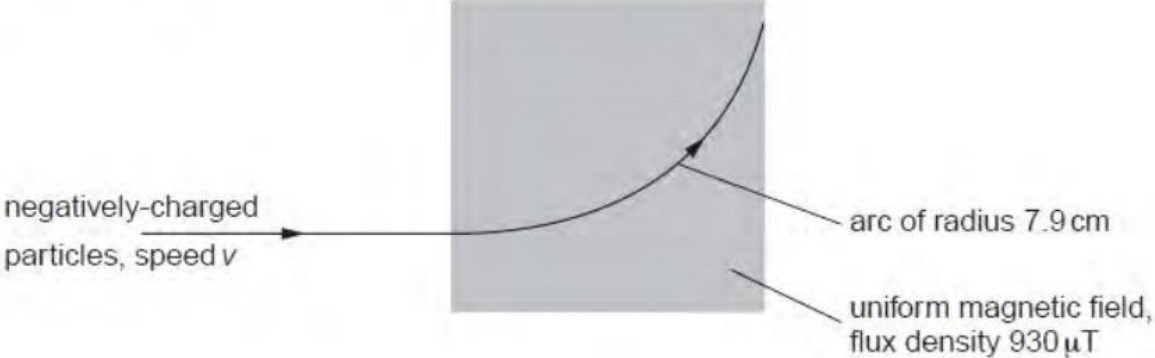


5	(a)	Define <i>magnetic flux density</i>	[1]
		(magnetic) force per unit length per unit current experienced by a (long straight) current-carrying conductor placed perpendicular to the (magnetic) field [B1] Comments: Many students did not remember this definition.	
	(b)	Negatively-charged particles are moving with speed v through a vacuum in a parallel beam. The particles enter a region of uniform magnetic field of flux density $930\ \mu\text{T}$. Initially, the particles are travelling at right-angles to the magnetic field. The path of a single particle is shown in Fig. 5.1.  Fig. 5.1 The negatively-charged particles follow a curved path of radius 7.9 cm in the magnetic field. A uniform electric field is then applied in the same region as the magnetic field. For an electric field strength of $12\ \text{kV m}^{-1}$, the particles pass through the region of the fields without deviation.	
	(i)	On Fig. 5.1, mark with an arrow the direction of the electric field.	[1]
		arrow pointing up the page [B1] Comments: Mostly well done.	
	(ii)	Calculate the speed v .	

			$v = \dots\dots\dots \text{ m s}^{-1}$	[3]
			$Eq = Bqv$ [C1] $v = (12 \times 10^3) / (930 \times 10^{-6})$ [C1] $= 1.3 \times 10^7 \text{ m s}^{-1}$ [A1] Comments: Mostly well done.	
		(iii)	Calculate the $\frac{\text{charge}}{\text{mass}}$ ratio of the negatively-charged particles.	
			$\text{ratio} = \dots\dots\dots \text{ C kg}^{-1}$	[3]
			$Bqv = mv^2 / r$ $q/m = (1.3 \times 10^7) / (7.9 \times 10^{-2} \times 930 \times 10^{-6})$ $= 1.8 \times 10^{11} \text{ C kg}^{-1}$ Most students got this correct.	
		[Total: 8]		

